

High-performance and complex materials

We rely on high-performance materials for many applications, and that reliance will only increase

- Maximum efficiency with minimum material use, high tailorability
- Urgent challenges in climate adaptation, durable infrastructure, sustainable structures

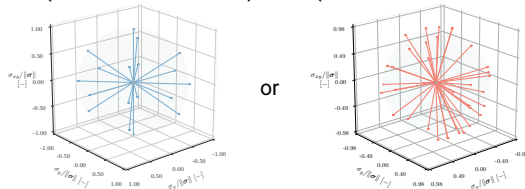


PRNNs versus conventional ML models

Testing with more challenging paths:

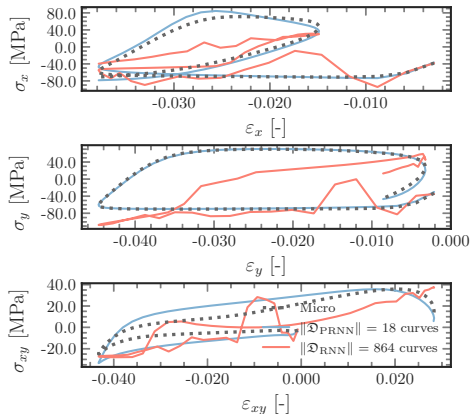
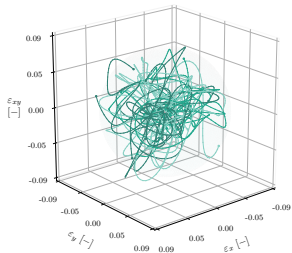
PRNN (18 monotonic curves) RNN (864 curves with unloading)

Train



or

Test



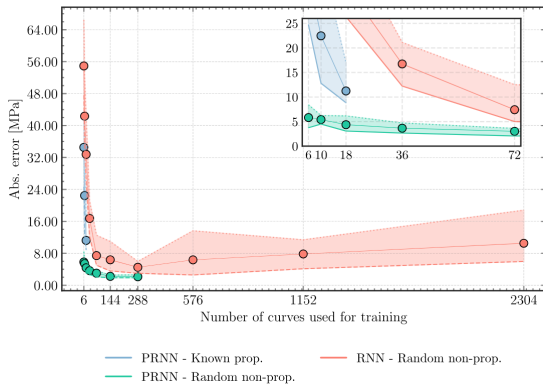
[Maia et al.(2023), CMAME]

PRNNs versus conventional ML models

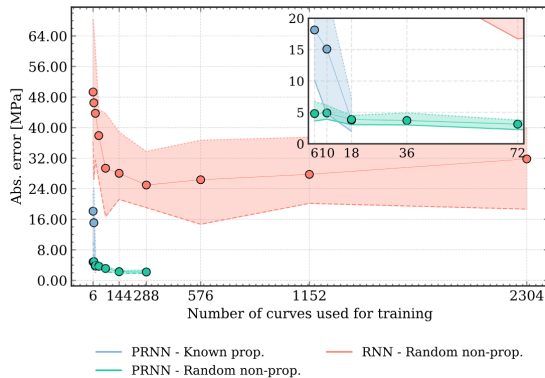
Learning curves with 10 realizations per dataset size:

- PRNNs remain accurate even when trained with very small datasets

Test set with 100 GP paths

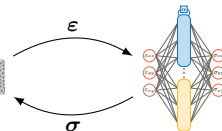
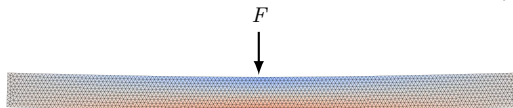
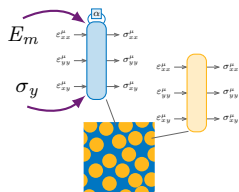


Test set with 100 monotonic paths ($10\times$ smaller Δt)



[Maia et al.(2023), CMAME]

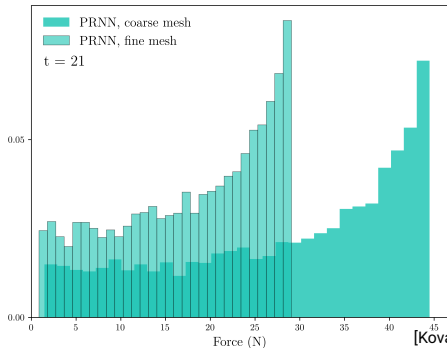
What we are now able to do: Multiscale UQ



Three-point bending, **fine** macro mesh:

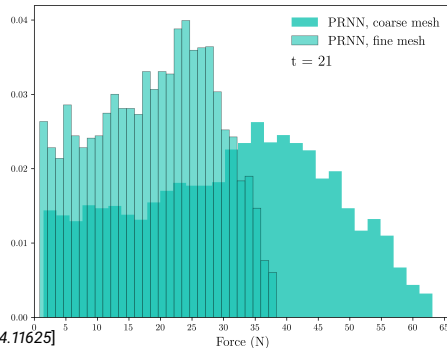
- Monte Carlo forward UQ, $N = 5000$ samples
- PRNN trained with a single (E_m, σ_y) pair
- Sequential FE^2 UQ would now take ≈ 50 years to run

$$p(F) = \int p(F|E_m)p(E_m) dE_m$$



[Kovacs et al.(2025), arXiv:2504.11625]

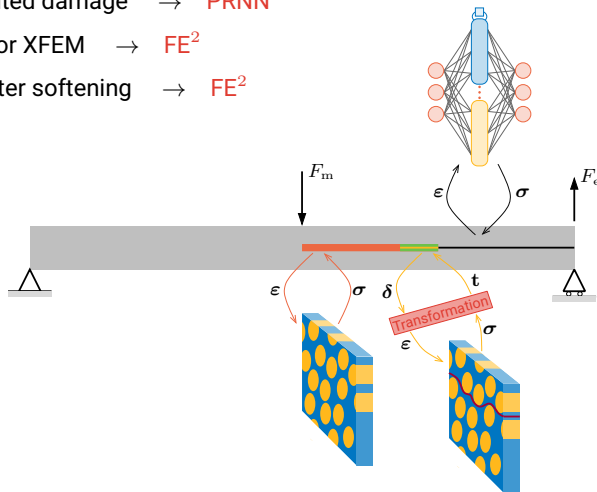
$$p(F) = \int p(F|E_m, \sigma_y)p(E_m, \sigma_y) dE_m d\sigma_y$$



But wait, what about failure? — localized damage

Multiscale modeling a mixed mode bending test

- Plasticity and distributed damage → PRNN
- Softening detection for XFEM → FE^2
- Cohesive tractions after softening → FE^2



[Ke et al., (In preparation)]